Courier Management System (CMS)

Project Documentation

TEAM MEMBERS:

JASWANTH ABINAV. S -TEAM LEADER

HARISANTHOSH. P

HEMACHANDRU. S

HEMANATH. J

NAAN MUDHALVAN DEPARTMENT- STAFF

SENTHILKUMAR. D- HOD

JAYAKUMAR. R-TUTIOR

1. INTRODUCTION

1.1 Project Overview:

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies.

The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

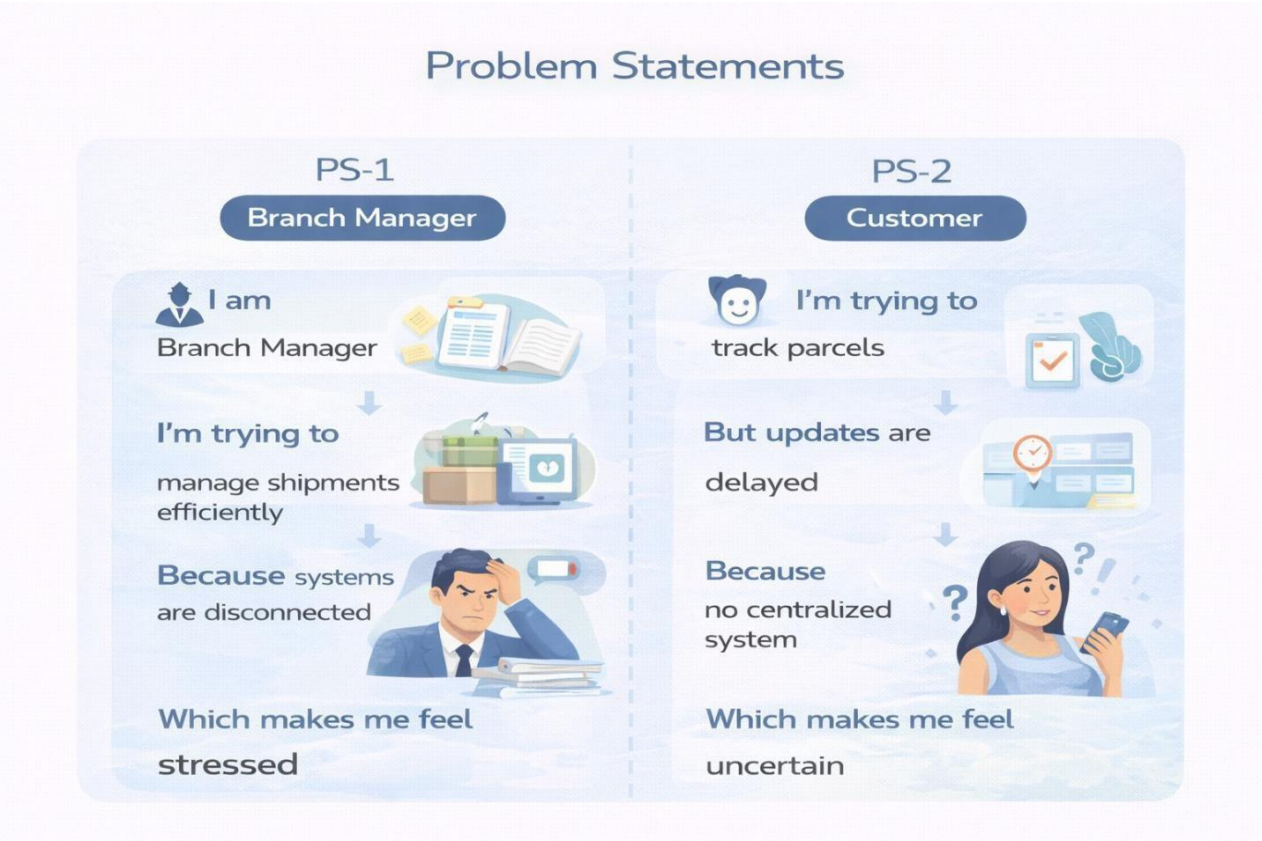
1.2 Purpose:

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

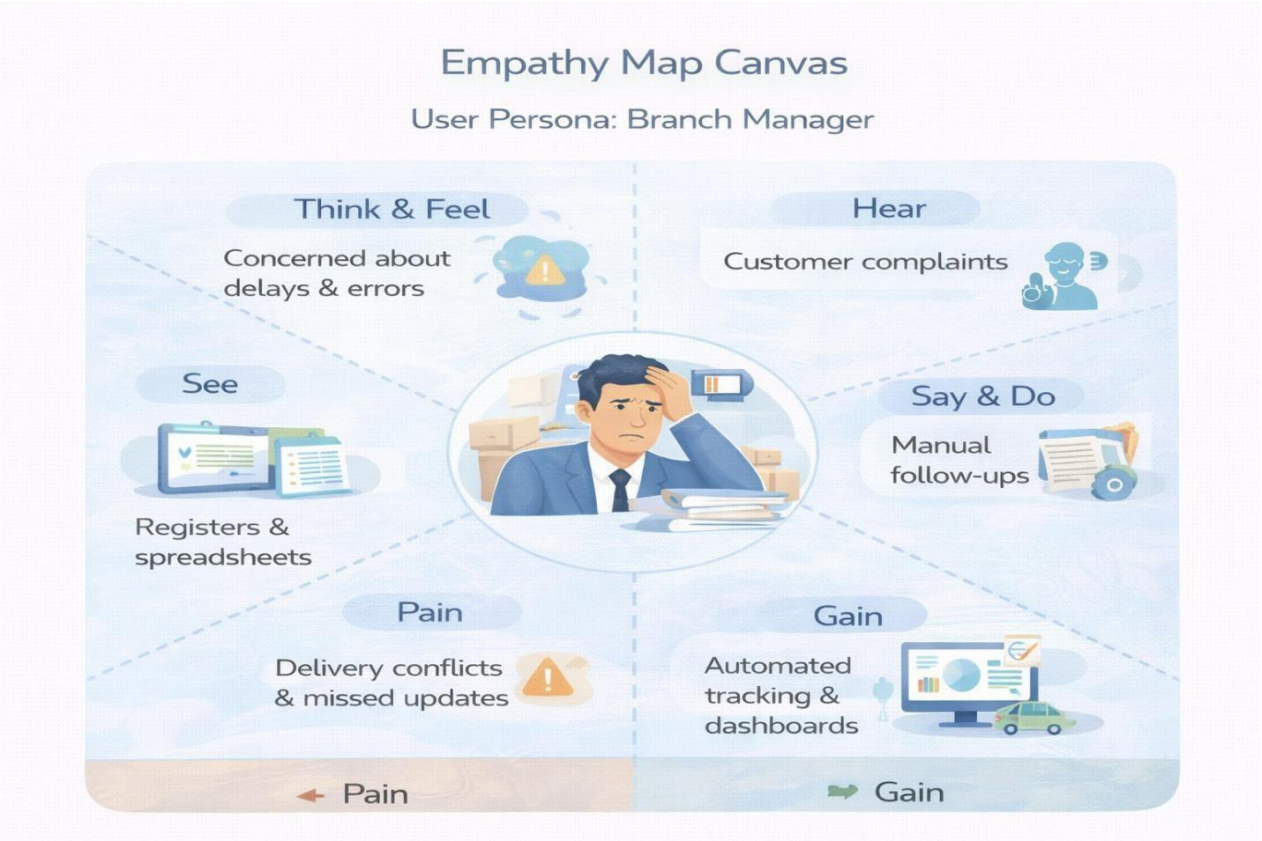
PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnected	stressed
PS-2	Customer	track parcels	updates are delayed	No centralized system	uncertain



2.2 Empathy Map Canvas

User Persona: Branch Manager

- **Think & Feel:** Concerned about delays & errors
- **Hear:** Customer complaints
- **See:** Registers & spreadsheets
- **Say & Do:** Manual follow-ups
- **Pain:** Delivery conflicts & missed updates
- **Gain:** Automated tracking & dashboards



2.3 Brainstorming & Idea Prioritization

Ideas Considered

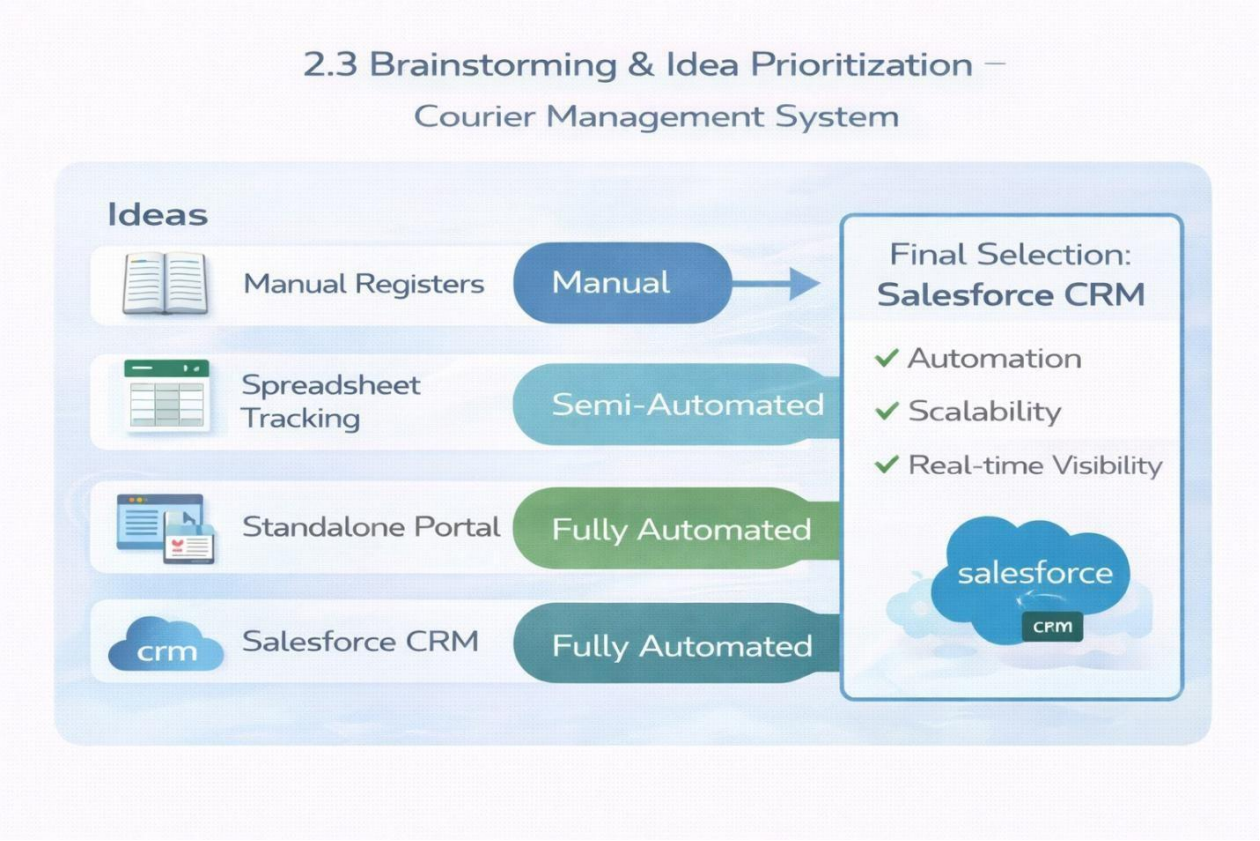
- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

Final Selection

Salesforce CRM selected due to:

- Automation

- Scalability
- Real-time visibility

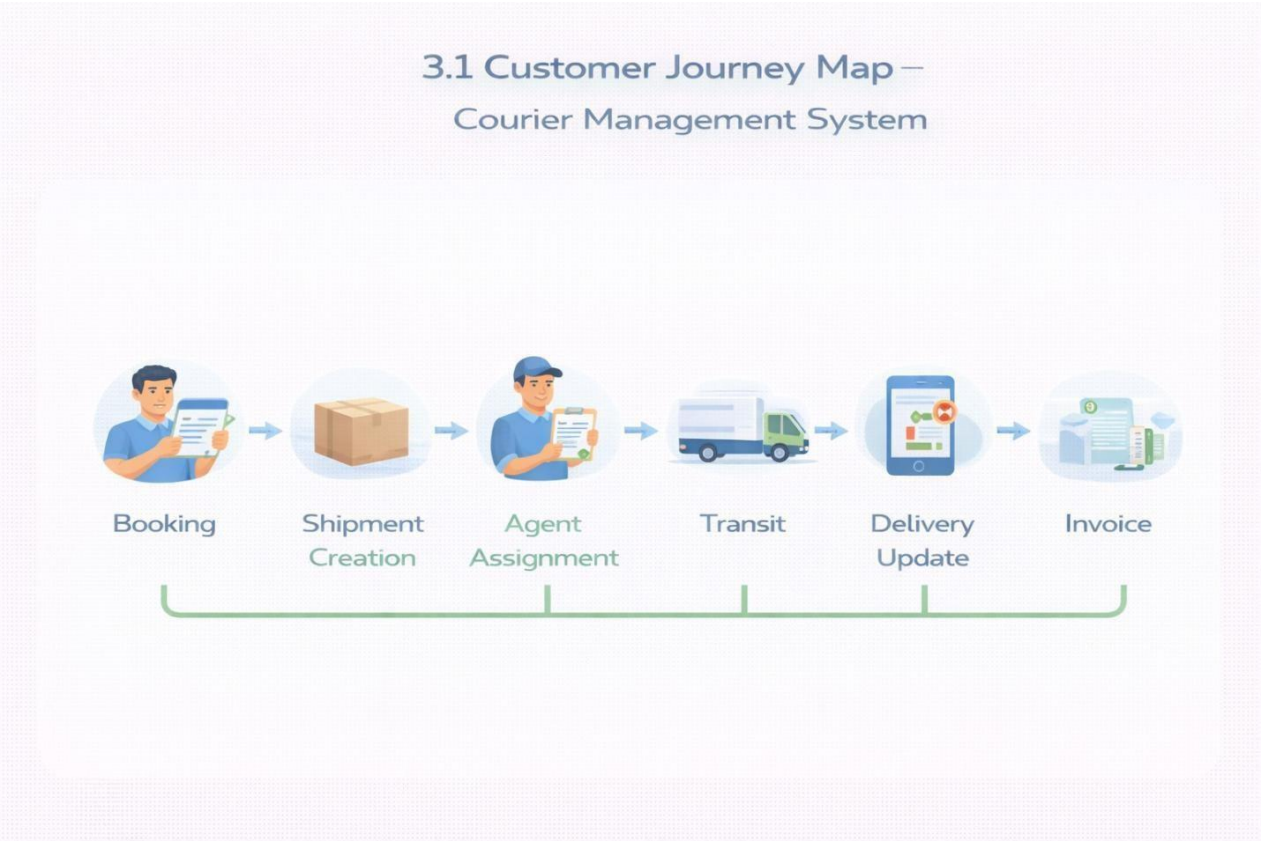


3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates

FR-5	Invoice generation
FR-6	Reports & dashboards

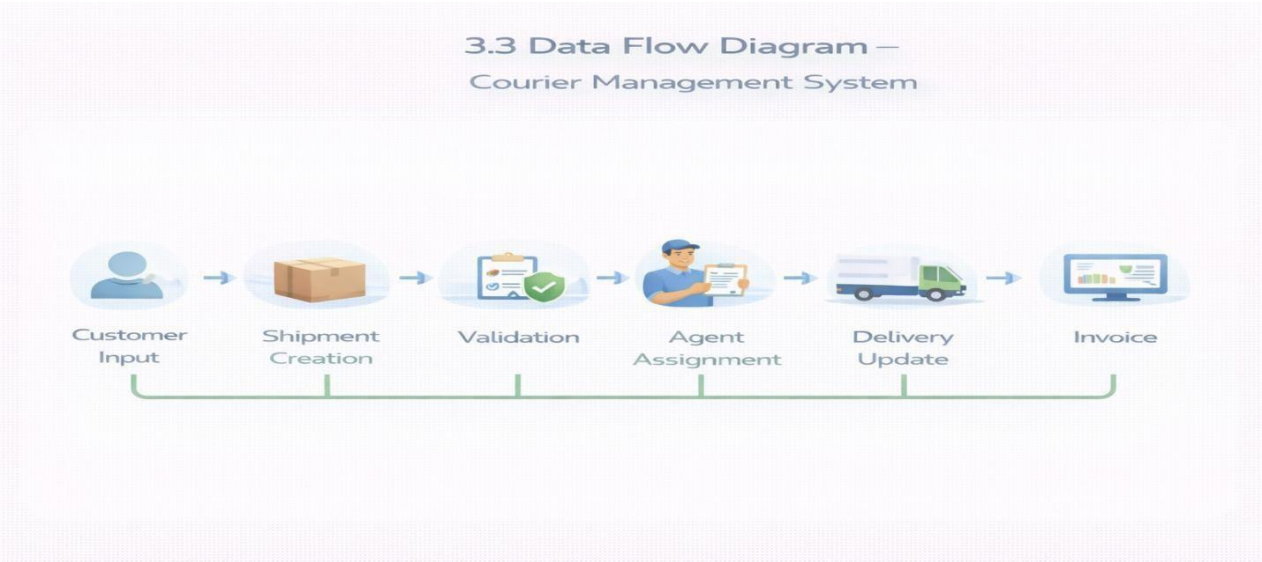
Non-Functional Requirements

NFR	**Description**
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports



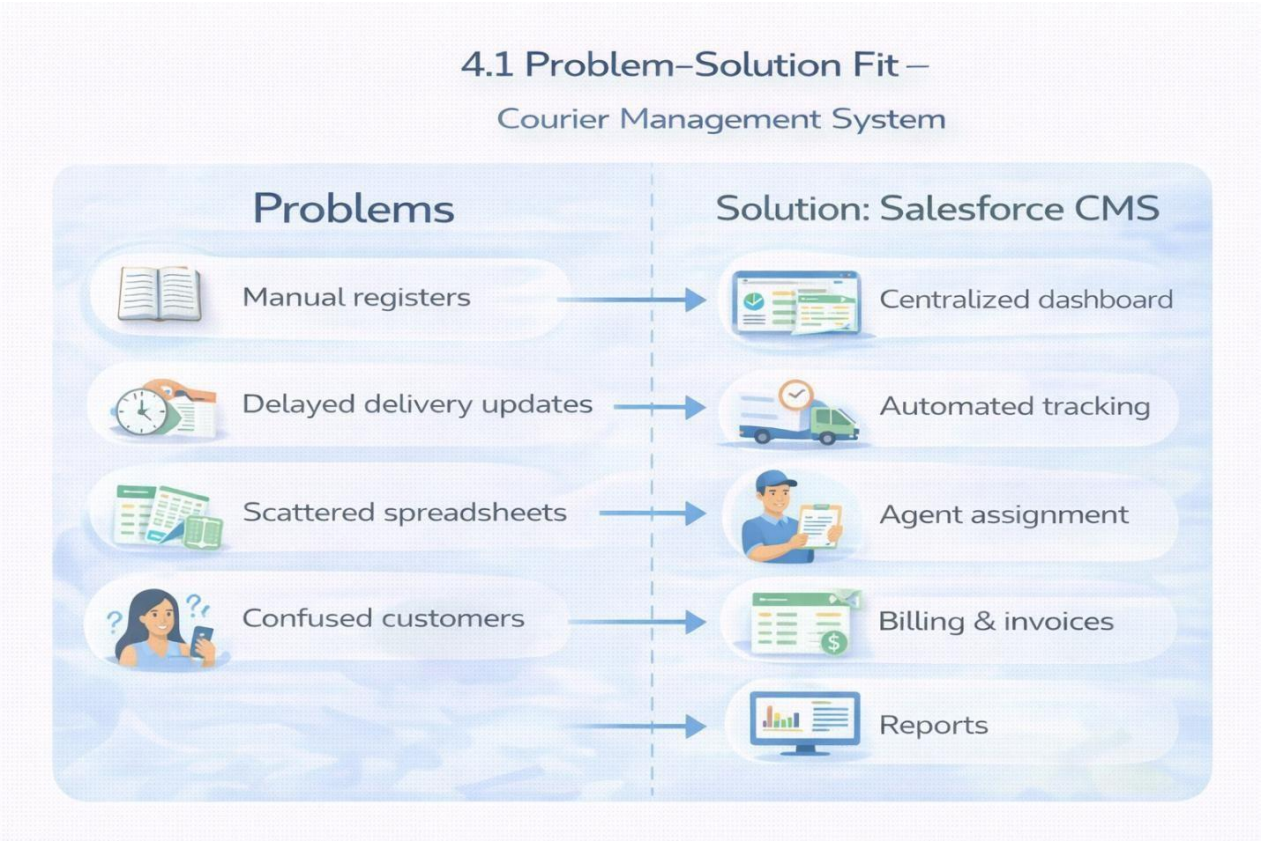
3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.



4.2 Proposed Solution

Functional Requirements

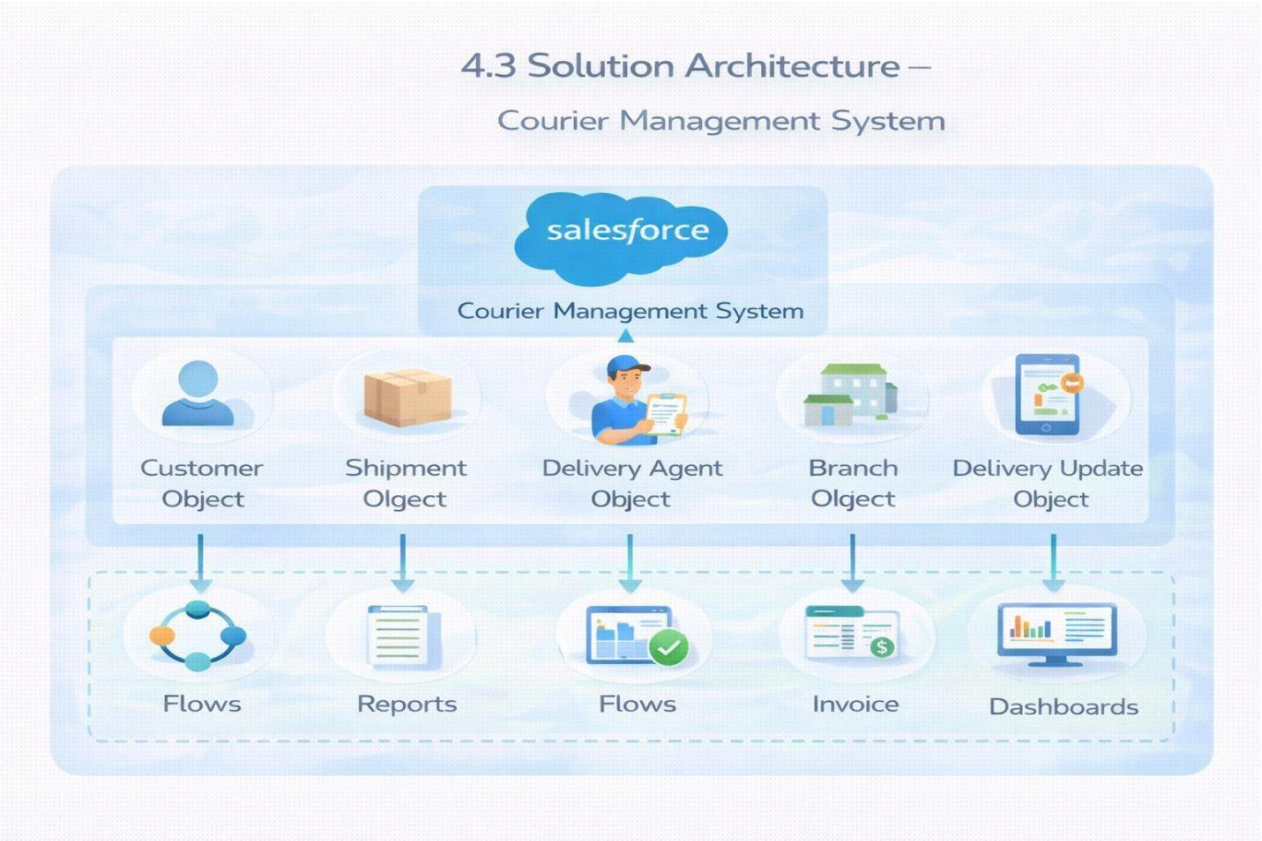
Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS

Innovation	Automated tracking & billing
Impact	Faster deliveries
Business Model	Courier services
Scalability	Multi-branch ready

4.3 Solution Architecture

Components:

- Customer Object ● Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

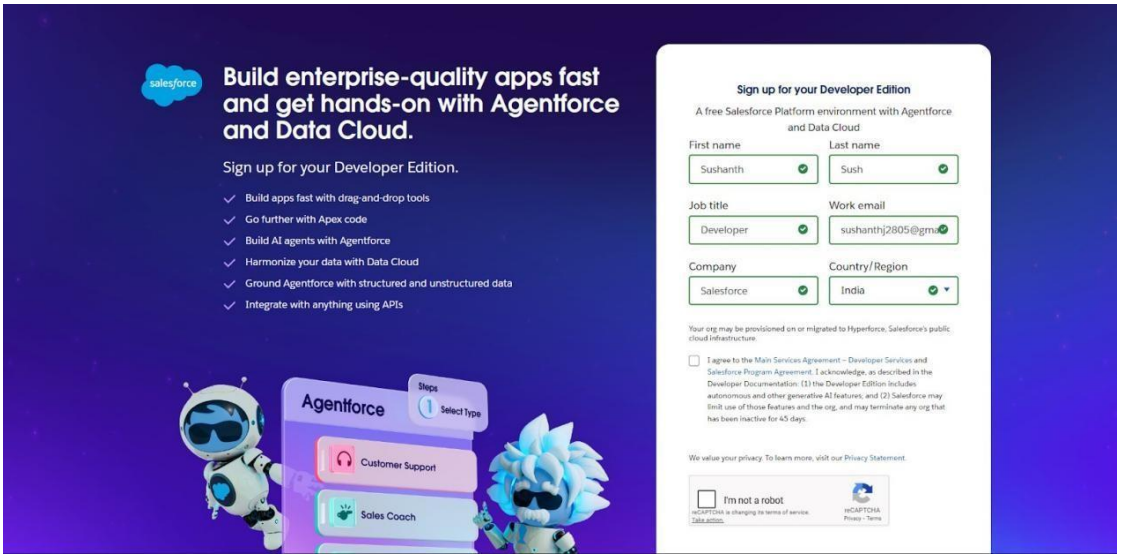
- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place.
Welcome to the Loan Management System project!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workflows—from borrower onboarding to loan approval, EMI tracking, and document verification.

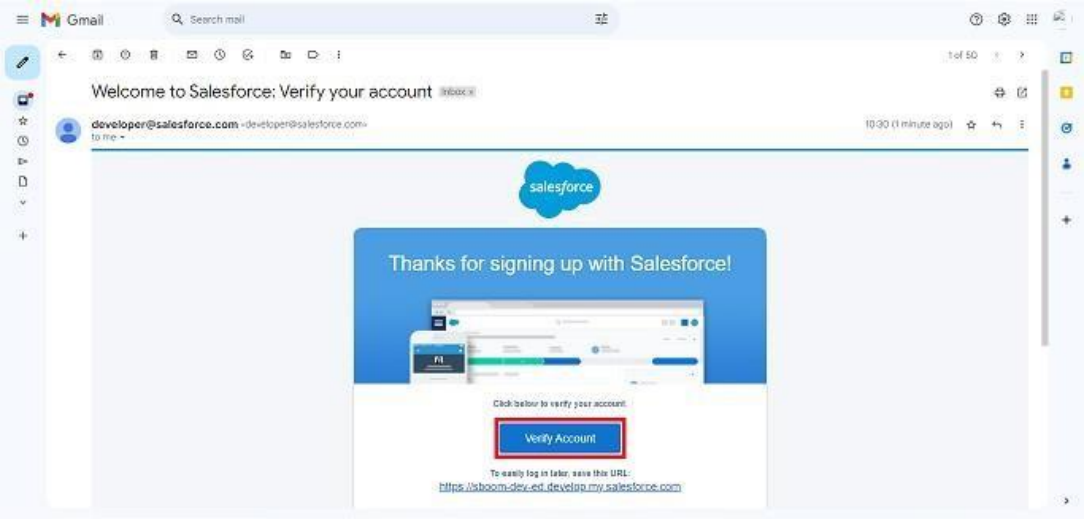


1) First name & Last name
2) Email
3) Job Title: Developer
4) Company: College Name
5) Country: India

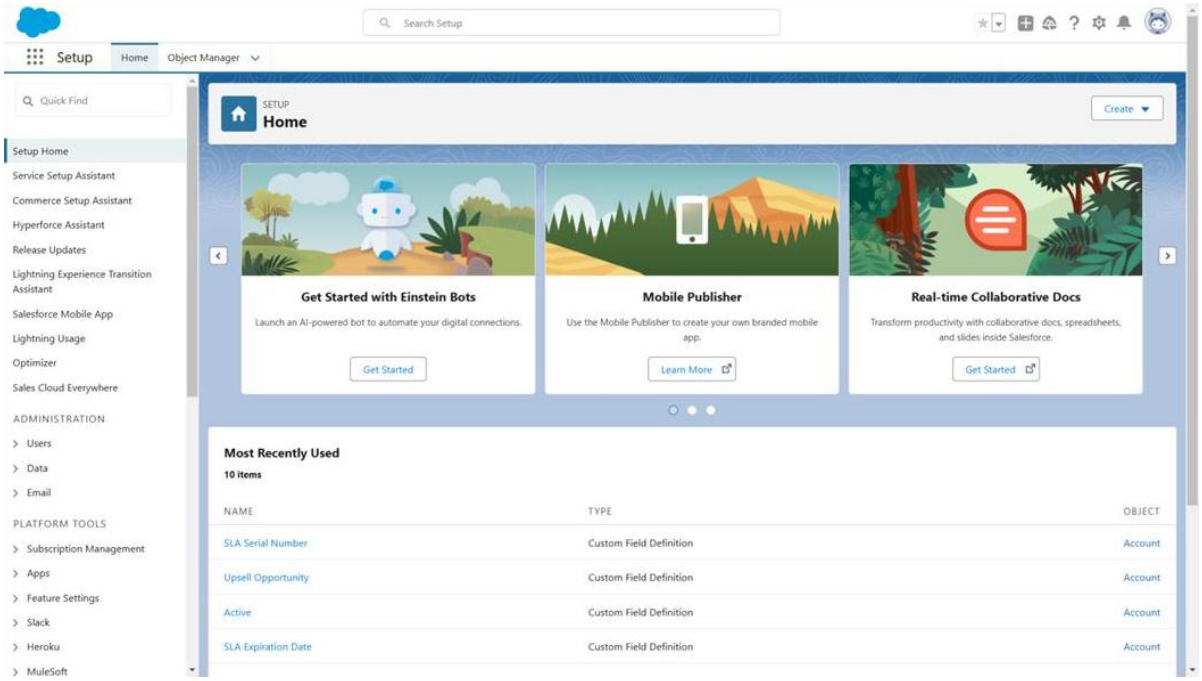
Click on sign me up after filling these.

Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.
4. Then you will redirect to your salesforce setup page.



Object:

Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information
2. **Shipment__c** – manages parcel details, source, destination, and tracking information
3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details
4. **Branch__c** – represents courier branch offices
5. **Delivery_Update__c** – stores status updates for each shipment
6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

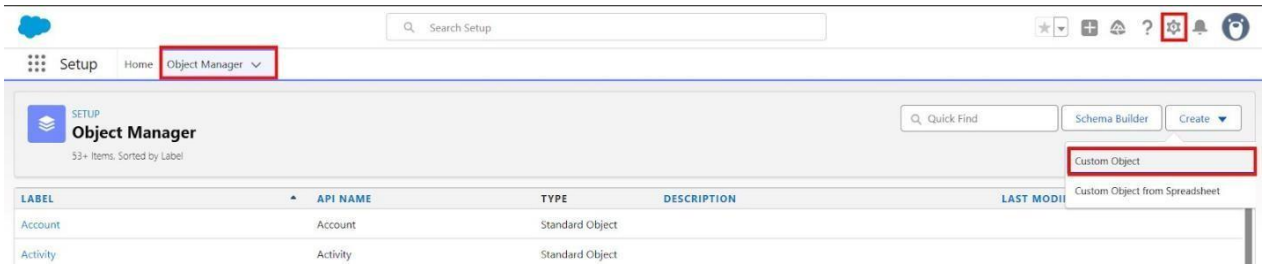
Delivery_Agent → Branch (Lookup)

Delivery_Update → Shipment (Master-Detail)

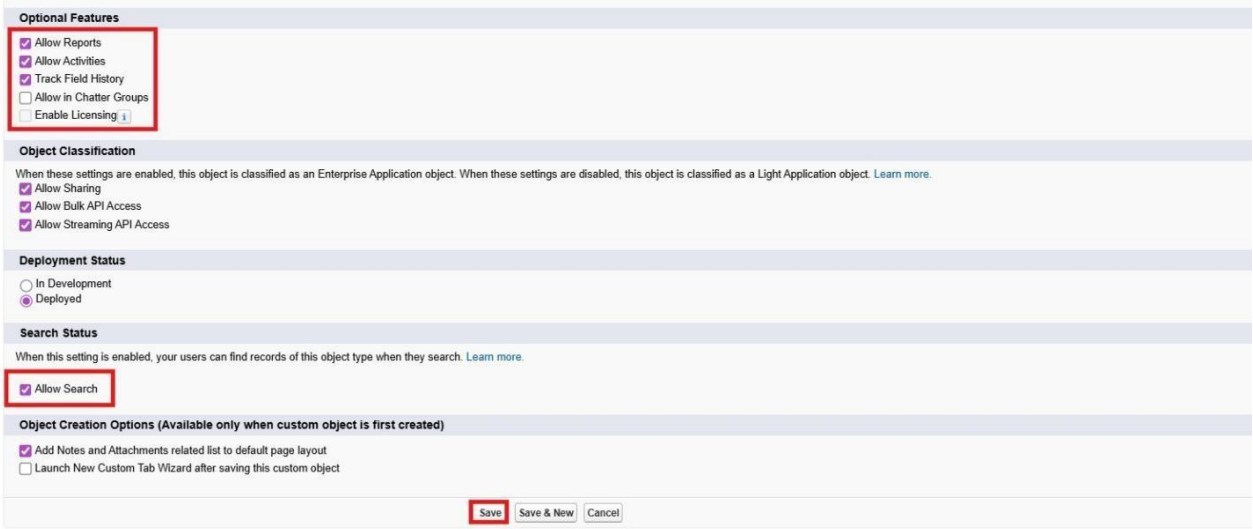
Invoice → Shipment (Lookup)

Steps to Implement:

Create Customer Object



1. Enter the following details:



Create Shipment Object

1. Go to Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Shipment

- Plural Label: Shipments

- Record Name: Tracking Number (Auto Number)

3. Example Format: SHP-{00000}

   Starting Number: 0

4. Enable:

   Allow Reports

- Track Field History

5. Click Save

●

Create Delivery_Agent Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Delivery Agent

- Plural Label: Delivery Agents

- Record Name: Agent Name (Text)

3. Enable:

- Allow Reports

- Track Field History

4. Click Save

Create Branch Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Branch

  Plural Label: Branches

- Record Name: Branch Name (Text)

3. Enable:

- Allow Reports

4. Click Save.

Create Delivery_Update Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Delivery Update

- Plural Label: Delivery Updates

- Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}
   Enter Starting number : 0
4. Enable:

- Allow Reports

- Track Field History

5. Click Save.

Create Invoice Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Invoice
- Plural Label: Invoices
- Record Name: Invoice Number (Auto Number)

3. Example Format: INV-{0000}
   Enter Starting number: 0
4. Enable:

- Allow Reports

Create Invoice Object

3. Setup → Object Manager → Create → Custom Object

4. Enter:

- Label: Invoice

- Plural Label: Invoices

- Record Name: Invoice Number (Auto Number)

5. Example Format: INV-{0000}
   Enter Starting number: 0
6. Enable:

- Allow Reports

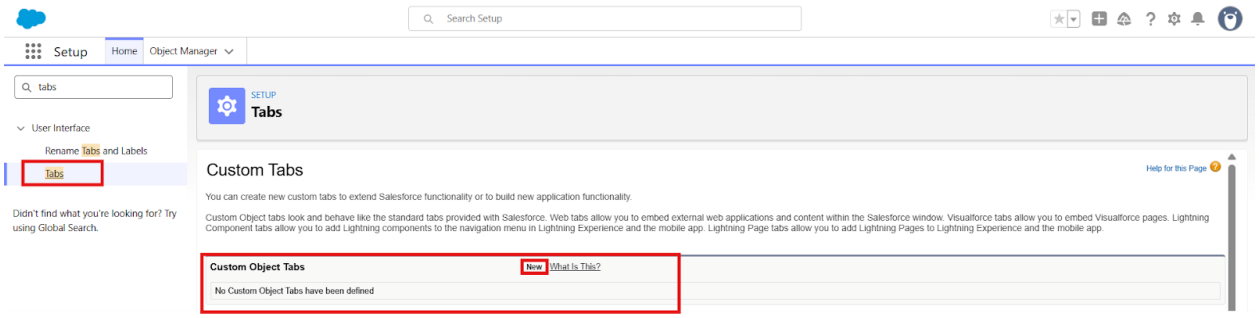
- Track Field History

5. Click Save

Tabs

Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.



3. For Object, select Customer.
4. For Tab Style, choose any icon.
5. Leave all default settings as they are.
6. Click Next, Next, and Save.

Creating a Tab for Shipment Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Shipment.
4. Choose any Tab Style.
5. Keep defaults unchanged.
6. Click Next, Next, and Save.

Creating a Tab for Delivery Agent Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Agent.
4. Pick any Tab Style.

5. Leave default settings.

6. Click Next, Next, and Save.

Creating a Tab for Branch Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

2. Under Custom Object Tabs, click New.

3. For Object, select Branch.

4. Choose any Tab Style.

5. Keep all defaults.

6. Click Next, Next, and Save.

Creating a Tab for Delivery Update Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.

3. For Object, select Delivery Update.

4. Select any Tab Style.

5. Keep default configurations.

6. Click Next, Next, and Save.

Creating a Tab for Invoice Object

1. Click Home, type Tabs in Quick Find, and select Tabs.

2. Under Custom Object Tabs, click New.

3. For Object, select Invoice.

4. Select any Tab Style.

5. Leave defaults unchanged.

6. Click Next, Next, and Save.

Fields & Relationships

Creation of fields for the Customer Object

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of customer
Email	Email__c	Email	Yes	Customer email
Phone	Phone__c	Phone	Yes	Customer contact number
Address	Address__c	Text Area	Yes	Customer address
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state
Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

Creation of fields for the Shipment Object

Field Name	API Name	Data Type	Required	Description
Tracking Number	Name	Auto Number	Yes	Unique ID
Customer	Customer__c	Lookup(Customer)	Yes	Related customer
Source Address	Source_Address__c	Text Area	Yes	Pickup location
Destination Address	Destination_Address__c	Text Area	Yes	Delivery location
Weight	Weight__c	Number	Yes	Parcel weight
Price	Price__c	Currency	Yes	Shipping charge
Status	Status__c	Picklist	Yes	Shipment status
Expected Date	Expected_Date__c	Date	Yes	Delivery estimate
Assigned Agent	Assigned_Agent__c	Lookup(Delivery_Agent)	No	Delivery agent
Shipment Type	Shipment_Type__c	Picklist	Yes	Domestic/International

Creation of fields for Delivery_Agent Object

Field Name	API Name	Data Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

Creation of fields for Branch Object

Field Name	API Name	Data Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

Creation of fields for Delivery_Update Object

Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes
Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	DateTime	Yes
Remarks	Remarks__c	Text	No

Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Relationships :

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery_Agent	Lookup
Delivery_Agent	Branch__c	Branch	Lookup
Delivery_Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Validation Rules:

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Create Validation Rules on Customer__c

1. Validation Rule: Email Must Not Be Blank

Steps:

- Click the Gear icon → Setup
- Go to Object Manager
- Search and select Customer__c
- From the left menu, click Validation Rules
- Click New
- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(Email__c)

- Click Check Syntax
- Error Message: Email is required.
- Error Location: Field → Email
- Click Save.

2. Validation Rule: Phone Must Be 10 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Phone_Number
- Check Active

Formula:

LEN(Phone _ c) <> 10

- Check Syntax
- Error Message: **Phone number must be 10 digits.**
- Error Location: Field → Phone
- Save

3. Validation Rule: Pincode Must Be 6 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Pincode
- Active → Yes
-

Formula:

LEN(Pincode _ c) <> 6

- Error Message: **Pincode must be 6 digits.**
- Location: Field → Pincode
- Save

Create Validation Rules on Shipment__c

1. Validation Rule: Weight Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Click Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

Formula:

Weight_ c <= 0

- Check Syntax

Error Message: **Weight must be greater than zero.**

- Location: Field → Weight
- Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Price
- Active → Yes

Formula:

Price _c <= 0

- Error Message: **Price must be greater than zero.**
- Location: Field → Price
- Save

3. Validation Rule: Expected Date Cannot Be in the Past

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Expected_Date

Formula:
Expected_Date__c < TODAY()

- Check Syntax
- Error: **Expected delivery date cannot be in the past.**
- Location: Field → Expected Date
- Save

Create Validation Rules on Delivery_Agent__c

1. Validation Rule: Email Must Be Valid

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Validation Rules → New
- Rule Name: Validate_Agent_Email
- Active → Yes

Formula:

NOT(CONTAINS(Email__c, "@"))

- Error Message: **Enter a valid email address.**
- Location: Field → Email

  Save

Create Validation Rules on Branch__c

1. Validation Rule: Branch Code Must Not Be Blank

Steps:

- Setup → Object Manager → Branch__c
- Validation Rules → New
- Rule Name: Branch_Code_Required

Formula:

ISBLANK(Branch_Code__c)

- Error Message: **Branch Code is mandatory.**
- Location: Field → Branch Code
- Save

Create Validation Rules on Delivery_Update__c

1. Validation Rule: Remarks Required If Status is Delayed

Steps:

- Setup → Object Manager → Delivery_Update__c
- Validation Rules → New
- Rule Name: Remarks_Required_For_Delay

Formula:

```
AND(
 ISPICKVAL(Status__c, "Delayed"),
 ISBLANK(Remarks__c)
)
```

- Error Message: **Remarks are required when status is Delayed.**
- Location: Field → Remarks
- Save

Page Layouts:

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.

They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.

By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

Edit Page Layout for Customer__c

Steps:

- Setup → Object Manager → Customer__c
- Click Page Layouts
- Click the default layout (usually Customer Layout)

Sections to Add & Arrange:

- Open the default layout

  In the top palette, click "Section"

- Drag the Section and drop it anywhere on the layout

A pop-up appears → Fill details:

- Section Name: "Customer Information"
- Layout:
  - 1-Column or
  - 2-Column (recommended)
- Tab-Key Order: Top-Down
- Click OK

The new section will appear on the layout.

- Drop fields inside the section (it will highlight).
- Repeat the same for each field.

Section 1: Customer Information

- Customer Name

- Customer Type
- Email
- Phone
- Is Active

Section 2: Address Details

- Address
- City
- State
- Pincode

Edit the layout as needed

Click Save.

Edit Page Layout for Shipment__c

Steps:

- Setup → Object Manager → Shipment__c
- Click Page Layouts
- Open the default Shipment Layout

Sections to Add:

Section 1: Shipment Details

- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

Section 2: Address Information

- Source Address
- Destination Address

Section 3: Delivery Information

- Expected Date
- Assigned Agent
- Arrange fields properly
- Click Save

Edit Page Layout for Delivery_Agent__c

Steps:

- Setup → Object Manager → Delivery_Agent__c
- Page Layouts
- Open default Delivery Agent Layout

Arrange Fields:

Agent Details

- Agent Name
- Email
- Phone
- Branch
- Status

  Joining Date

Additional Information

- Related Shipments
- Related Delivery Update

Edit Page Layout for Branch__c

Steps:

- Setup → Object Manager → Branch__c
- Page Layouts
- Open default Branch Layout

-

Sections:

Branch Information

- Branch Name
- Branch Code
- City
- State

Manager Name

- Arrange fields
- Click Save

Edit Page Layout for Delivery_Update__c

Steps:

- Setup → Object Manager → Delivery_Update__c
- Page Layouts
- Open default Delivery Update Layout

Sections:

Update Information

- Update Number
- Shipment

- Status
- Update Date
- Remarks
- Arrange properly
- Click **Save**

Edit Page Layout for Invoice__c

Steps:

Setup → Object Manager → Invoice__c

- Page Layouts
- Open default Invoice Layout

Sections:

Invoice Details

- Invoice Number
- Shipment
- Amount
- Invoice Date
- Payment Status
- Arrange fields as needed
- Click Save

Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

Auto-Update Shipment Status When Delivery Update is Created

Scenario:

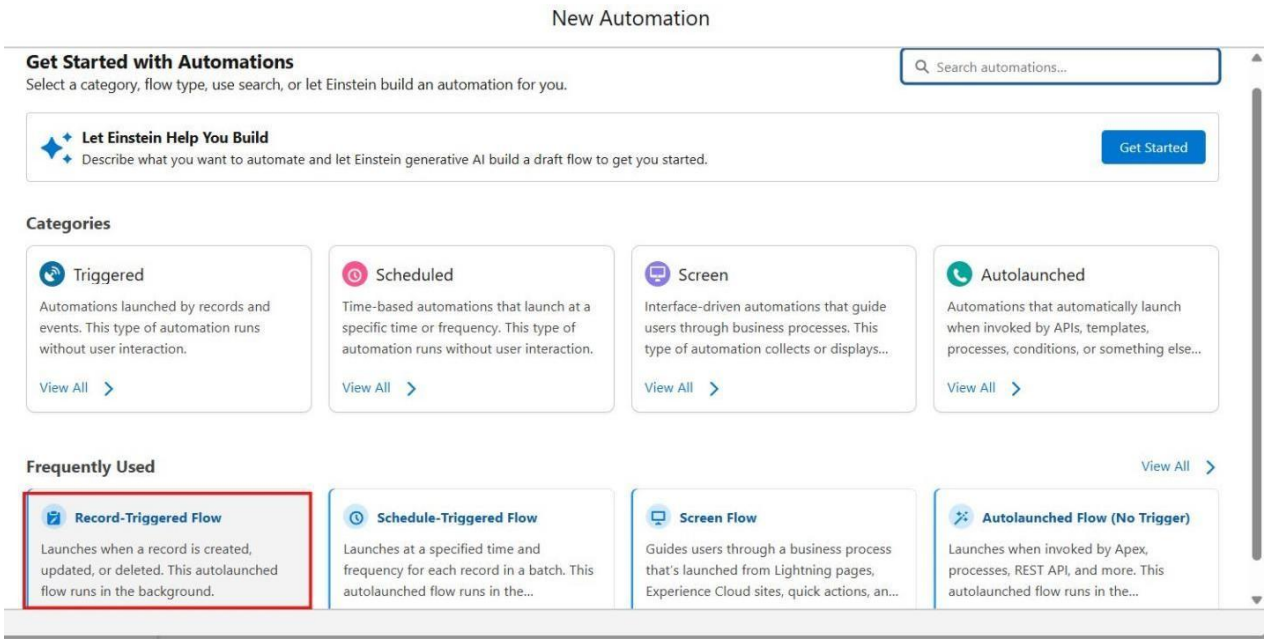
Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update.

Part 1: Create the Record-Triggered Flow

Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows
- Click the New Flow button

- Select Record-Triggered Flow
- Click Create



Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: select Delivery_Update__c
- Trigger the Flow When:
    - Select A record is created
- Condition Requirements:
    - Choose: None (Always Run the Flow)
    - Optimize the Flow For
    - Select Actions and Related Records
        - Click Done

-

Part 2: Update Related Shipment Record
Step 3: Add "Update Records" Element

- Click the + icon under the Start element
- Select Update Records

Configure the Update Records Element:

- Label: Update_Shipment_Status
- How to Find Records to Update:
    - Select Specify conditions to identify records

Set Filter Conditions:

- Object: Shipment__c
- Field: Id
- Operator: Equals
- Value: {!$Record.Shipment__c}

(This means update the Shipment linked to the Delivery Update)

Set Field Values to Update Add field mapping:

Shipment Field	Value Source
Status__c	{!$Record.Status__c}

(This copies status from Delivery Update to Shipment)

- Click Done

Part 3: Save and

Activate Step 4:

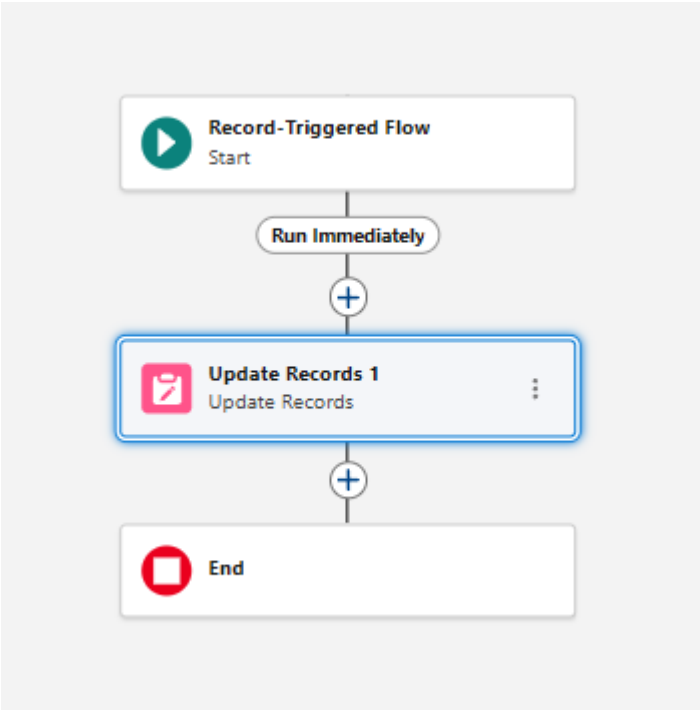
Save Flow

- Click Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- Click Save

Step 5: Activate Flow

- Click Activate
- Flow status should show Active

Final Flow Layout:



The Lightning App :

What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Create a Lightning App

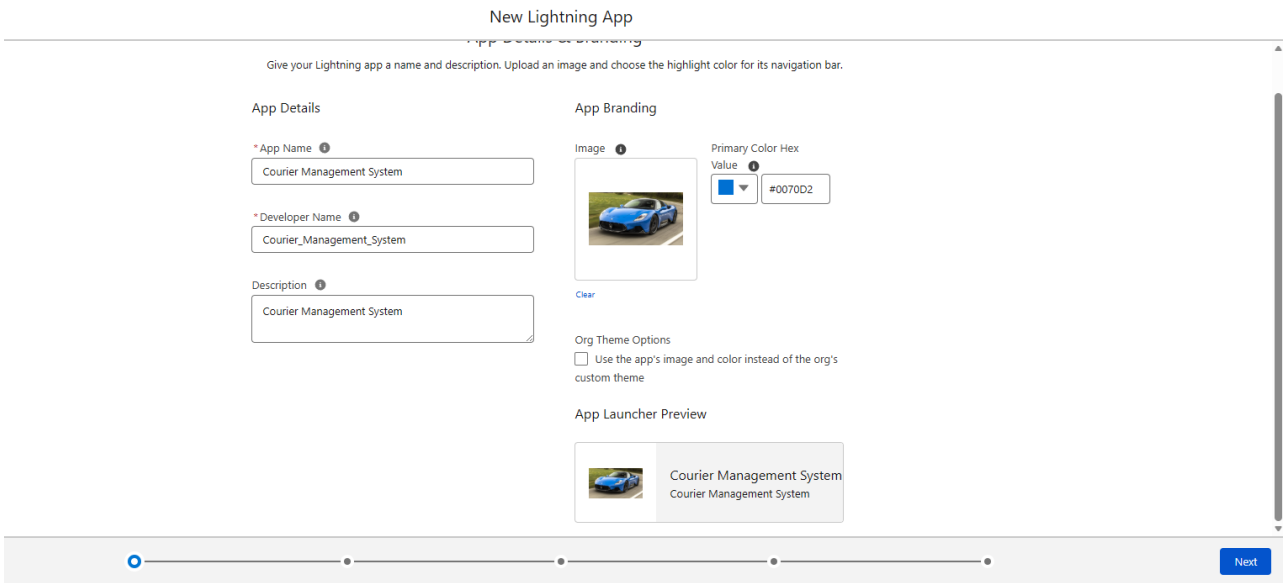
Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.

2. Click New Lightning App.

    - Enter the Courier Management System as the App Name.

3. Upload an image or icon for the app, then click Next.



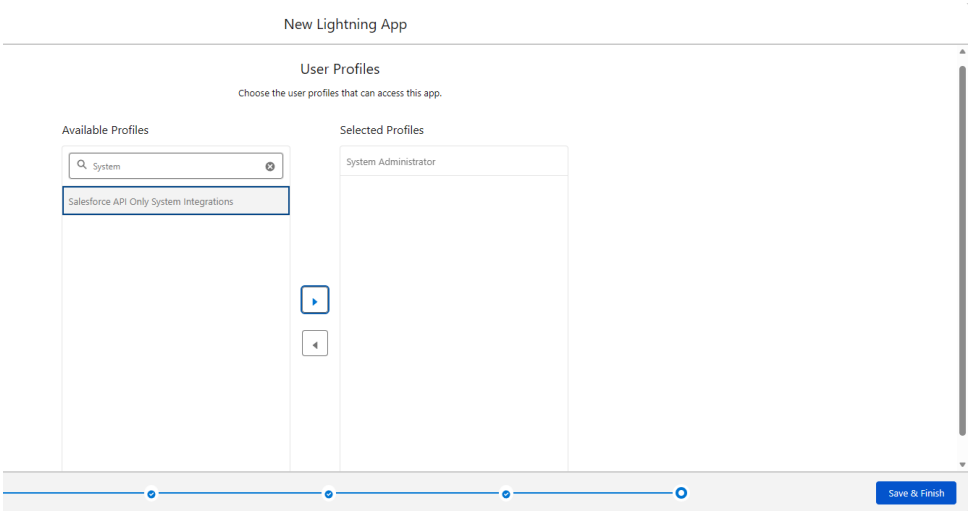
4. Under App Options, leave the default selections and click Next.

5. Under Utility Items, leave as it is and click Next.

6. From Available Items, select the following and move them to Selected Items:

- Customer
- Shipment
- Delivery Agent
- Branch
- Delivery Update
- Invoice
- Reports
- Dashboards

7. Click Next.
8. From Available Profiles, select System Administrator and move it to Selected Profiles.
9. Click Save & Finish.



Profile:

What is a profile?

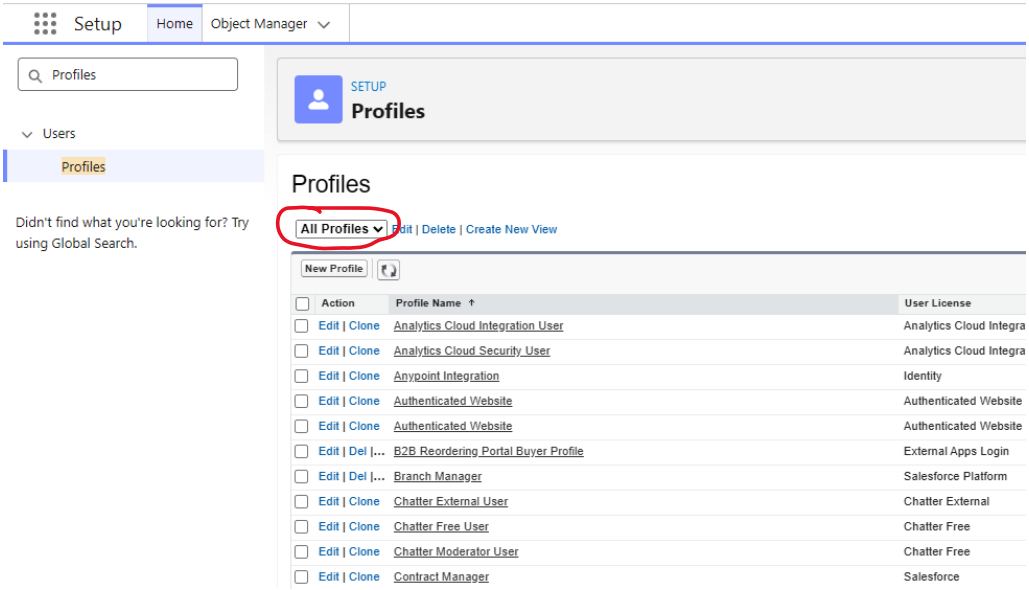
A profile is a group/collection of settings and permissions that define what a user can do in salesforce.

A profile can be assigned to many users, but users can be assigned a single profile at a time.

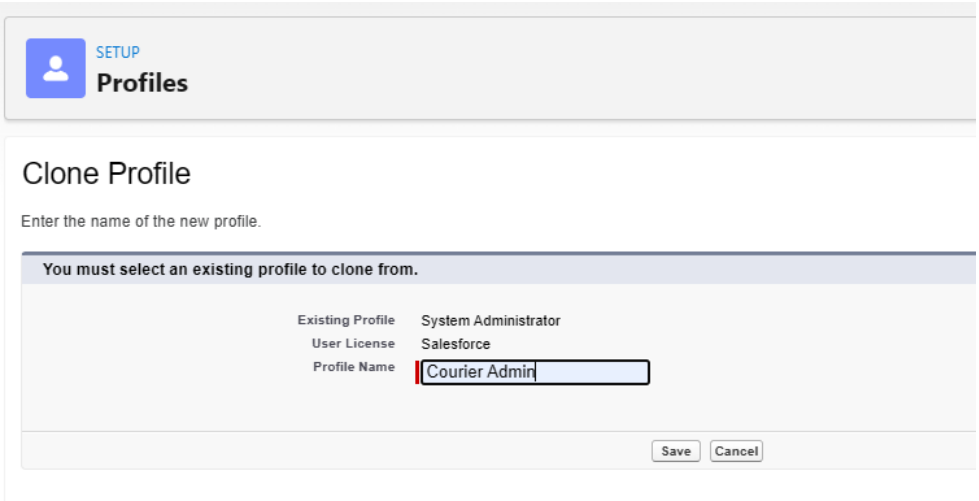
Create a Custom Profile – Courier Admin

Create a Custom Profile:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.



4. Select Existing Profile as: System Administrator.
5. Enter Profile Name: Courier Admin.



6. Click Save.

Why use System Administrator as Base?

It already has:

- Modify All Data
- Customize Application
- Full CRUD access on all objects

By cloning it, the **Courier Admin profile automatically inherits full access**. You can later restrict permissions if required.

Create a Custom Profile – Branch Manager

Steps:

1. From Setup, enter Profiles in the Quick Find box.

2. Click Profiles.
3. Click New Profile.
4. Select Existing Profile as: Standard Platform User.
5. Enter Profile Name: Branch Manager.
6. Click Save.
7. Click Edit on the Branch Manager profile.

Set App & Tab Settings

Under Assigned Apps / Custom App Settings, set the default app to:

Courier Management System App

Ensure Visible Tabs Include:

Set these tabs as Default On or Visible:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

Custom Object Permissions

Scroll to Custom Object Permissions and set access as follows:

Object	Permissions to Grant
Customer__c	Create, Read, Edit
Shipment__c	Create, Read, Edit
Delivery_Agent__c	Read, Edit
Branch__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Invoice__c	Read

	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records	Modify All Records	View A Fields
Customer__c	☑	☑	☑	☑	☐	☐	☐
Shipment__c	☑	☑	☑	☑	☐	☐	☐
Delivery_Agent__c	☑	☑	☑	☐	☐	☐	☐
Branch__c	☑	☑	☑	☐	☐	☐	☐
Delivery_Update__c	☑	☑	☑	☑	☐	☐	☐
Invoice__c	☑	☐					

Create a Custom Profile – Delivery Agent

Object Permissions :

Object	Permissions
Shipment__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Customer__c	Read
Invoice__c	Read

Create a Custom Profile – Customer Support

Object Permissions :

Object	Access
Customer__c	Read, Edit
Shipment__c	Read, Edit
Delivery_Update__c	Read
Invoice__c	Read

Final Profiles Created in Project :

Profile Name	Purpose
Courier Admin	Full system control
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking

Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization.

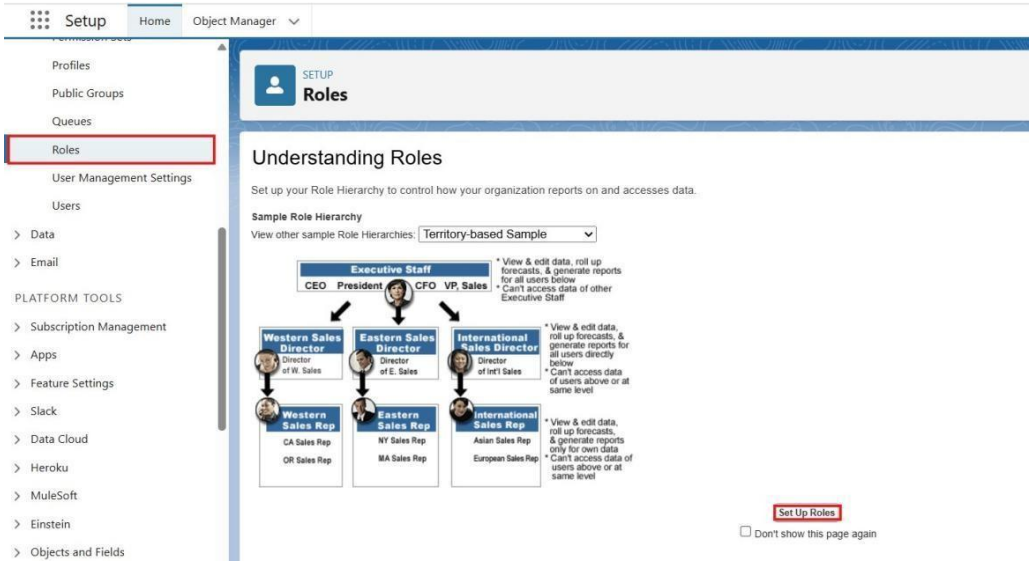
Users with higher roles have greater access to data and more control over the system.

Creation of Courier Admin Role

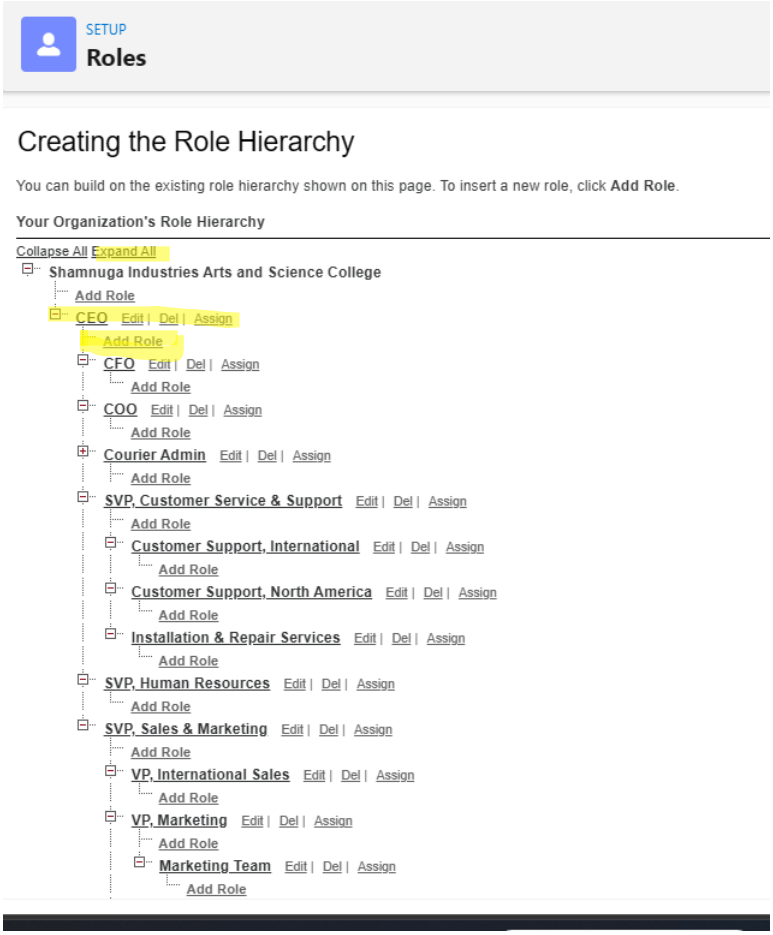
Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.

2. Select Set Up Roles.



3. Click on Expand All and click Add Role under CEO.



4. Give Label as “Courier Admin”.
    - The Role Name will auto-populate.
5. Click Save.

Creating Branch Manager Role Under Courier Admin Role

Create another role:

1. Create a role with the name “Branch Manager”.
2. Below the Courier Admin, click on Add Role.
3. Enter the label name as “Branch Manager” and the Role Name will automatically populate.
4. Click Save.

Creating Delivery Agent Role Under Branch Manager Role

Steps:

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.
    - Role Name will auto-populate.
3. Click Save.

Creating Customer Support Role Under Branch Manager

Steps:

1. Under the Branch Manager role, click on Add Role.

2. Enter Label as “Customer Support”.

3. Role Name will automatically populate.

4. Click Save.

Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

User :

What is a user?

A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

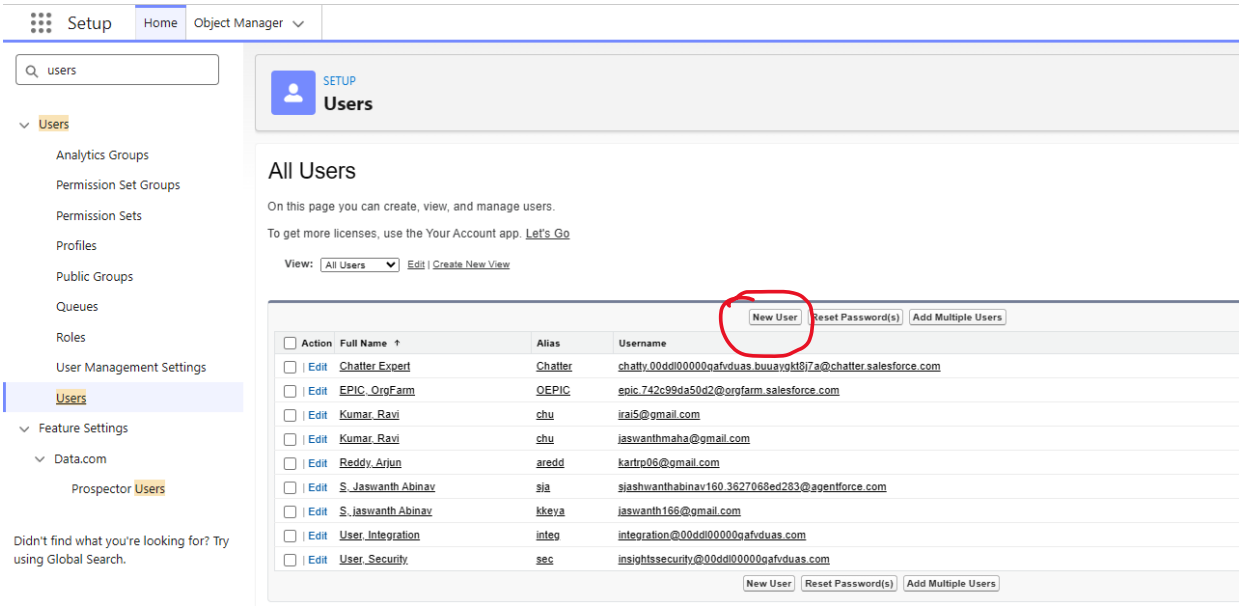
NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, So If you don’t find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

To Create a User

Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.

2. Click New User.



Fill in the Fields:

Field Name	Value
First Name	Enter first name (Example: Ravi)
Last Name	Enter last name (Example: Kumar)
Alias	Give an alias name
Email ID	Enter your personal email ID
Username	Must be in the format: text@text.text and must be unique
Nickname	Give a nickname
Role	Select **Courier Admin**

User License	Salesforce
Profile	**Courier Admin**

3. Click Save.

Creating Other Users

Repeat the same steps as **Activity 1** for creating additional users.

Use the following details:

User Type	Role	User License	Profile
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform	Delivery Agent
Customer Support	Customer Support	Salesforce Platform	Customer Support

Reports & Dashboards

Create Report Types

Step 1: Open Report Types

1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.

Step 2: Create Shipment Report Type

- Primary Object: Shipment__c

- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

Step 3: Create Customer Report Type

- Primary Object: Customer__c
- Label: Customer Summary Report
- Category: Reports

  Deployment Status: Deployed

  Click Next → Save

4: Create Agent Performance Report Type

Primary Object: Delivery_Agent__c

- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed

- Click Next → Save **Activity**

2: Create Reports

Report 1: Shipment Status Summary Report

Step 1: Create the Report

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click Continue **Step 2: Add Columns**

Add these fields:

- Tracking Number
- Customer
- Status

  Expected Date

  Assigned Agent

3: Add Filters

Show Me: All Shipments

- Date Filter: This Month
- Status Filter: Select specific Status (Optional)

Step 4: Group & Summarize

- Group rows by Status__c
- Add summary on Price

Step 5: Save Report

- Click Save & Run
- Name: Shipment Status Summary
- Folder: CMS Reports

Agent Performance Report

Step 1: Create Report

- New Report → Choose Agent Performance Report
- Click Continue

Step 2: Add Columns

- Agent Name
- Branch

  Number of Shipments

  Status

3: Group Data

Group by Delivery_Agent__c

Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports
- Save & Run

Report 3: Revenue Report

Steps

- Create report using Shipment Tracking Report Type

Add fields:

- Tracking Number
- Customer

Create Dashboard

Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create

Add Dashboard Components

Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

Agent Performance (Bar Chart)

- Select report: Agent Performance Report
- Chart Type: Bar Chart
- Click Add

Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

Save & Activate Dashboard

1. Click Save
2. Click Done

Share Dashboard With:

- Courier Admin
- Branch Managers
- Customer Support Team

Final Layout :

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

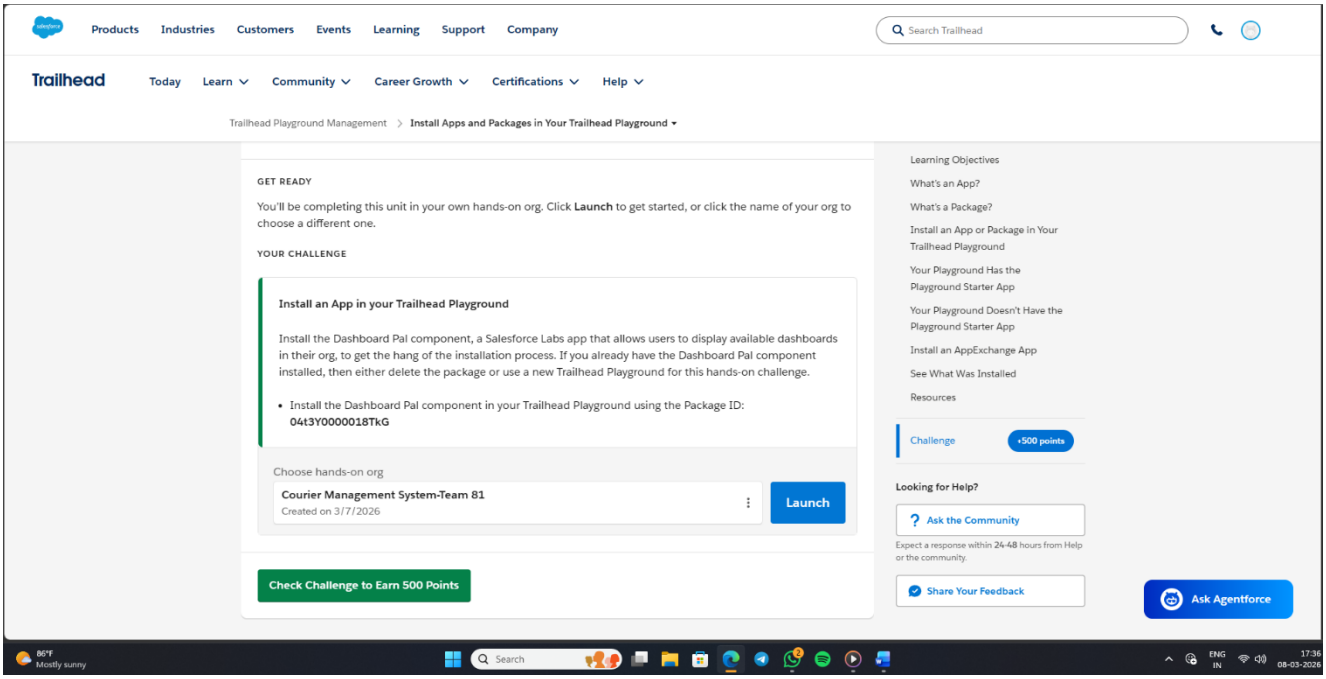
- Centralized data
- Automated workflows

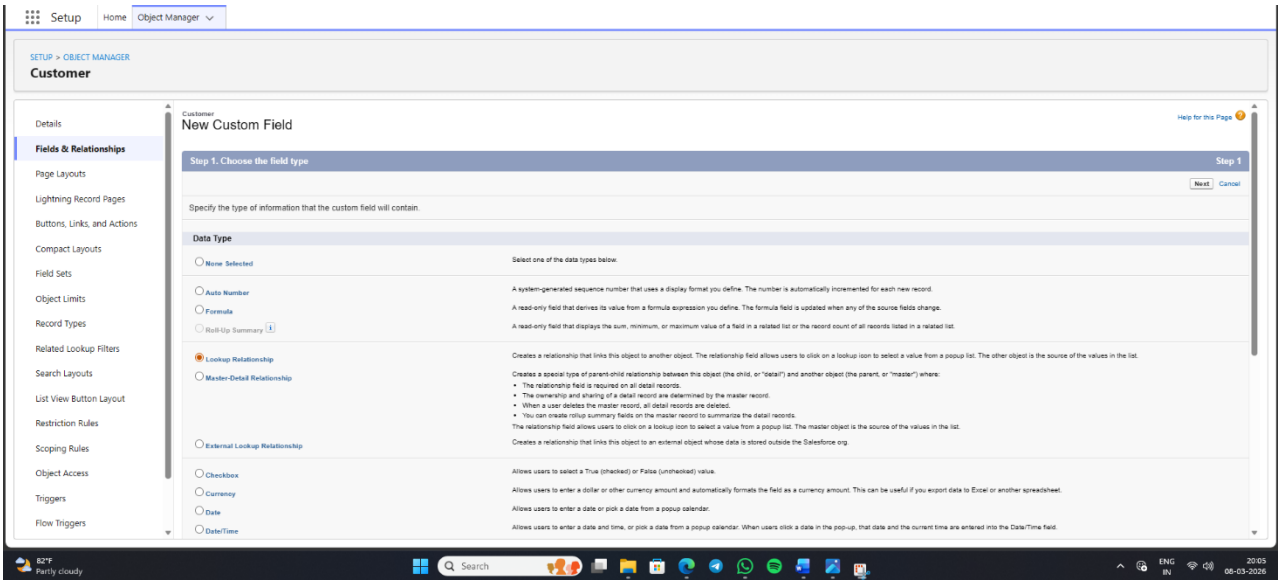
- Real-time reporting

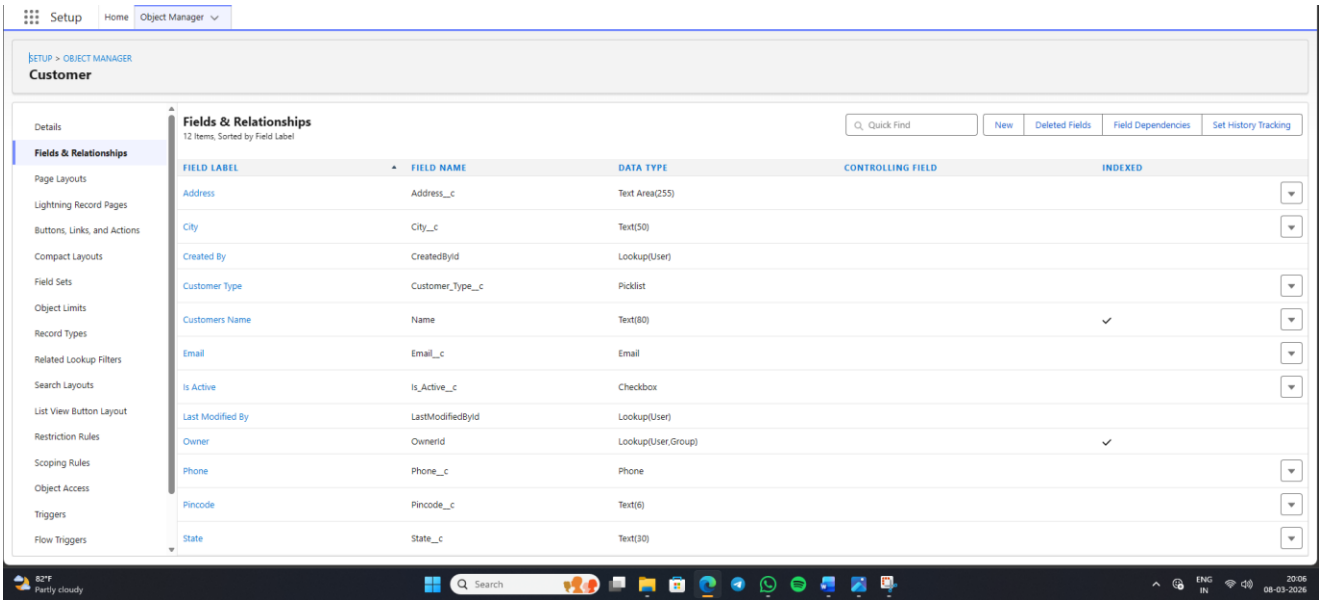
Disadvantages

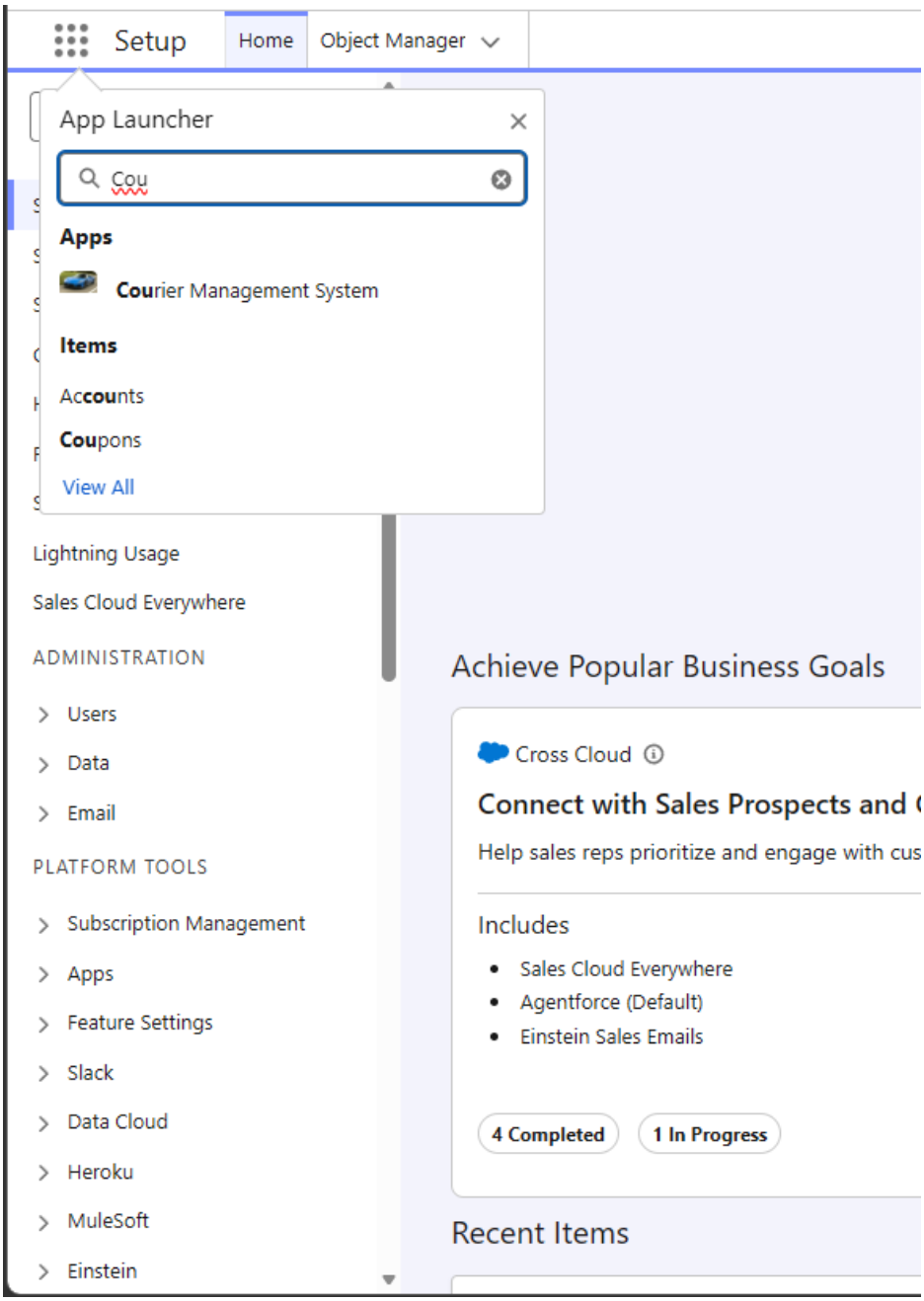
- Salesforce dependency
- Requires initial configuration effort

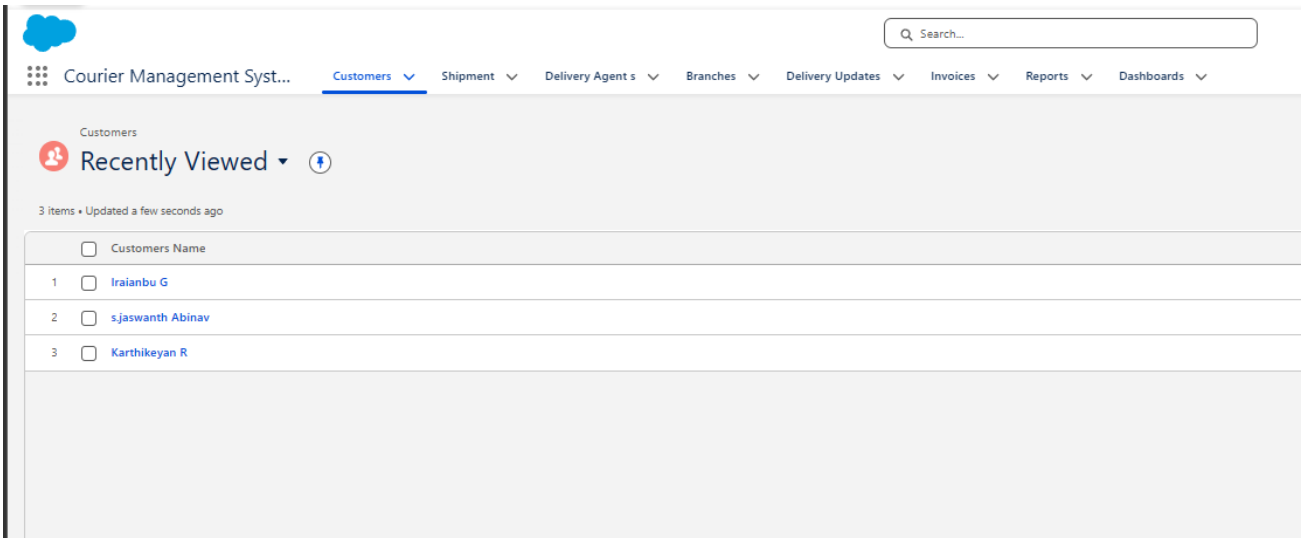
OUTPUT AND SCREENSHOTS

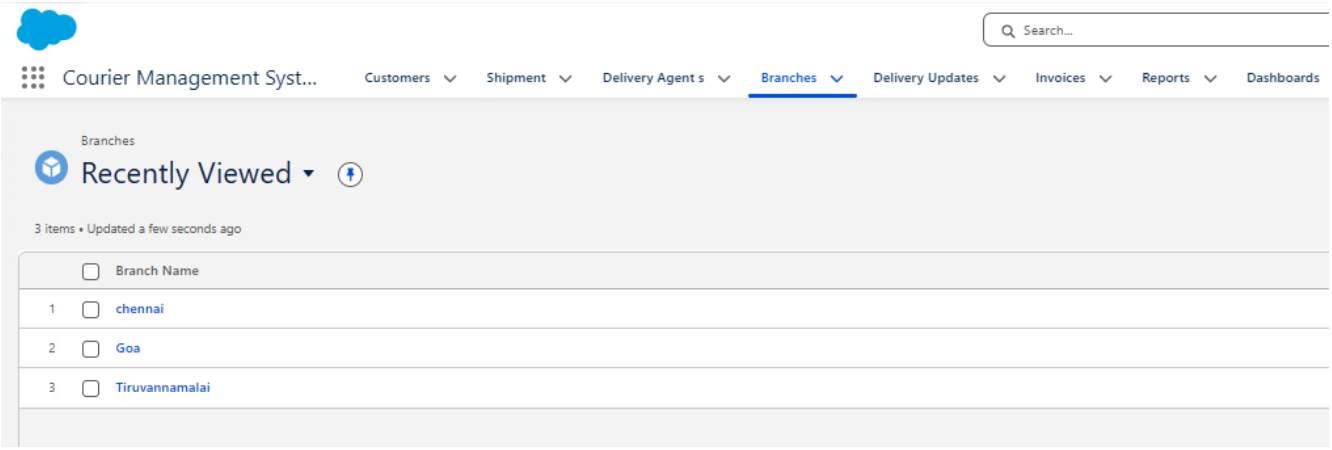


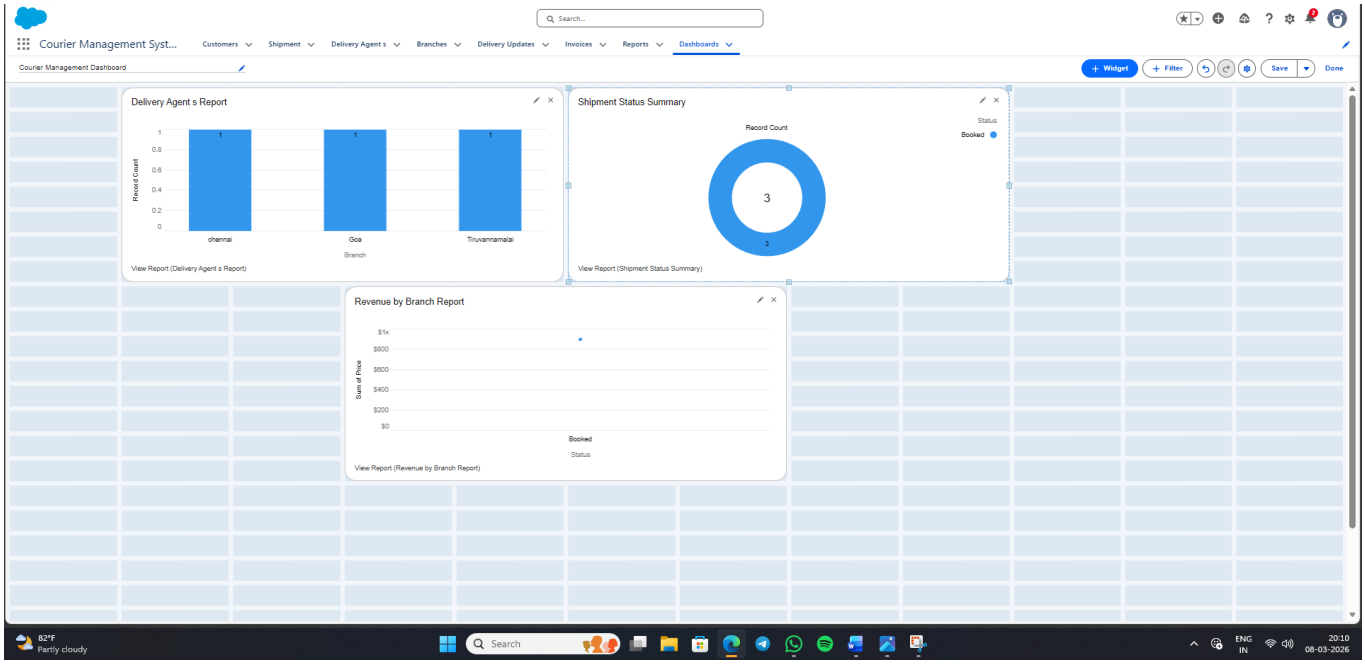

Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Customer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Customer
New Custom Field
Help for this Page
Step 1. Choose the field type
Step 1
Next
Cancel
Specify the type of information that the custom field will contain.
Data Type
None Selected
Select one of the data types below.
Auto Number
A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
Formula
A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
Roll-Up Summary
A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
Lookup Relationship
Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
Master-Detail Relationship
Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:
The relationship field is required on all detail records.
The ownership and sharing of a detail record are determined by the master record.
When a user deletes the master record, all detail records are deleted.
You can create rollup summary fields on the master record to summarize the detail records.
The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
External Lookup Relationship
Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.
Checkbox
Allows users to select a True (checked) or False (unchecked) value.
Currency
Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
Date
Allows users to enter a date or pick a date from a popup calendar.
DateTime
Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.
82°F
Partly cloudy
Search
ENG
IN
20:05
08-03-2026


Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Customer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Fields & Relationships
12 Items, Sorted by Field Label
Quick Find
New
Deleted Fields
Field Dependencies
Set History Tracking
FIELD LABEL
FIELD NAME
DATA TYPE
CONTROLLING FIELD
INDEXED
Address
Address__c
Text Area(255)
City
City__c
Text(50)
Created By
CreatedById
Lookup(User)
Customer Type
Customer_Type__c
Picklist
Customers Name
Name
Text(80)
Email
Email__c
Email
Is Active
Is_Active__c
Checkbox
Last Modified By
LastModifiedById
Lookup(User)
Owner
OwnerId
Lookup(User,Group)
Phone
Phone__c
Phone
Pincode
Pincode__c
Text(6)
State
State__c
Text(30)
82°F
Partly cloudy
Search
ENG
IN
20:06
08-03-2026


Setup
Home
Object Manager
App Launcher
Cou
Apps
Courier Management System
Items
Accounts
Coupons
View All
Lightning Usage
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
PLATFORM TOOLS
Subscription Management
Apps
Feature Settings
Slack
Data Cloud
Heroku
MuleSoft
Einstein
Achieve Popular Business Goals
Cross Cloud
Connect with Sales Prospects and
Help sales reps prioritize and engage with cus
Includes
Sales Cloud Everywhere
Agentforce (Default)
Einstein Sales Emails
4 Completed
1 In Progress
Recent Items


Search...
Courier Management Syst...
Customers
Shipment
Delivery Agent s
Branches
Delivery Updates
Invoices
Reports
Dashboards
Customers
Recently Viewed
3 items • Updated a few seconds ago
Customers Name
1 Iraianbu G
2 s.jaswanth Abinav
3 Karthikeyan R


Search...
Courier Management Syst...
Customers
Shipment
Delivery Agent s
Branches
Delivery Updates
Invoices
Reports
Dashboards
Branches
Recently Viewed
3 items • Updated a few seconds ago
Branch Name
1 chennai
2 Goa
3 Tiruvannamalai


Courier Management Syst...
Customers
Shipment
Delivery Agent s
Branches
Delivery Updates
Invoices
Reports
Dashboards
Courier Management Dashboard
Widget
Filter
Save
Done
Delivery Agent s Report
Record Count
chennai
Goa
Tiruvannamalai
Branch
View Report (Delivery Agent s Report)
Shipment Status Summary
Record Count
3
Status
Booked
View Report (Shipment Status Summary)
Revenue by Branch Report
Sum of Price
Booked
Status
View Report (Revenue by Branch Report)
82°F
Partly cloudy
Search
ENG
IN
20:10
08-03-2026

9. Conclusion

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization
- Artificial Intelligence & Machine Learning
- Real-Time Visibility & Tracking
- Sustainability & Green Logistics
- Hyperlocal & Quick Commerce
- Cloud & API Integrations

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You